MID-LEVEL HARMONIC AUDIO FEATURES FOR MUSICAL STYLE CLASSIFICATION

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ABSTRACT

The extraction of harmonic information from musical audio is fundamental for several music information retrieval tasks. In this paper, we propose novel harmonic audio features based on the perceptually-inspired tonal interval vector space, computed as the Fourier transform of chroma vectors. Our contribution includes mid-level features for musical dissonance, chromaticity, dyadicity, triadicity, diminished quality, diatonicity, and wholetoneness. Moreover, we quantify the perceptual relationship between short- and long-term harmonic structures, tonal dispersion, harmonic changes, and complexity. Beyond the computation on fixed-size windows, we propose a context-sensitive harmonic segmentation approach. We assess the robustness of the new harmonic features in style classification tasks regarding classical music periods and composers. Our results align with, slightly outperforming, existing features and suggest that other musical properties than those in state-of-the-art literature are partially captured. We discuss the features regarding their musical interpretation and compare the different feature groups regarding their effectiveness for discriminating classical music periods and composers.

1. INTRODUCTION

Over the last decades, music consumption has shifted from physical media to streaming services comprising large digital collections [1]. Methods for organizing these collections are fundamental for user navigation, browsing, and retrieval. In this context, a significant effort has been devoted to the automatic classification of musical audio signals into style or genre categories within Musical Information Retrieval (MIR) [2–4]. While the traditional approach to such tasks is based on hand-crafted features and classical machine learning, end-to-end deep-learning approaches have led to major improvements [4]. Nevertheless, strategies based on hand-crafted *mid-level* features are still of relevance since they allow interpretable and controllable systems that focus on specific aspects of the music.

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Existing approaches for style classification mostly rely on timbral or rhythmic mid-level features, which appear suitable for discriminating top-level genres such as pop, rock, jazz, and classical music [2, 5, 6] or sub-genres of popular music [7, 8]. However, such features are less suitable for discriminating sub-genres or historical periods within Western classical music—consider, e. g., the co-existence of solo piano music composed over several centuries [9]. To address this challenge, harmonic features have shown promising results [10–13]. Yet, existing harmonic audio features exhibit two main limitations. First, many of these features focus on low-level and short-term properties, which do not explicitly capture the horizontal or long-term structure of harmony, known to be relevant to style classification. Second, these features do not explicitly consider perceptual qualities, such as the degree of dissonance or the perceptual relationship of sonorities.

To account for the two limitations identified above, we consider in this paper a set of harmonic features based on the Tonal Interval Vector (TIV) space proposed in [14]. Multi-level pitch is mapped into a 6-dimensional complex space whose distances capture perceptual relationships between sonorities. We make the following four main contributions. (1) We consider the TIVs proposed in [14] for style classification. (2) On the TIV space, we advance a set of novel harmonic features for capturing long-term hierarchical harmonic relationships. (3) Moreover, we propose a structural audio segmentation based on harmonic changes and compare this approach to a fixed-window segmentation. (4) To assess the newly proposed harmonic features, we perform experiments for style classification of Western classical music, considering historical periods (eras) and composers as sub-genre taxonomies.

The paper is structured as follows. Section 2 discusses related work on the harmonic description of musical audio and style classification. Section 3 presents novel harmonic features based on the TIV space. Section 4 presents our experimental results using TIV features for style classification compared to the state-of-the-art. Finally, Section 5 presents the conclusions and future work.

2. RELATED WORK

The harmonic description of musical audio typically adopts chroma vectors to represent the energy of pitch-class content and to design higher-level harmonic features. Several methods have been proposed for the extraction of

chroma vectors [15–18]. The Non-Negative Least Squares (NNLS) chroma reduces the impact of overtones and has shown to be one of the most robust chroma vector representations for audio transcription [17].

For describing harmonic properties based on chroma vectors, Weiß et al. proposed a set of template-based chord and interval features [10] as well as tonal complexity features [11] for style period and composer classification within Western classical music. Most of these features are transposition-invariant and therefore do not depend on the key of the piece. These two sets of features are detailed in Sections 2.1 and 2.2. Furthermore, mid-level features for capturing chord transitions over time using Hidden Markov Models were proposed in [13].

2.1 Template-based Features

Motivated by the study on stylistic features from pitch-class sets by Honingh and Bod [19, 20], Weiß et al. [10] propose a set of template-based features (denoted as $\mathbf{F}$) studying the likelihood of a given complementary interval or triad type in a chroma vector $\mathbf{c} = (c_0, c_1, ..., c_{11}) \in \mathbb{R}^{12}$. For example, the likelihood of a perfect fourth/fifth interval, F_{IC5}, results from multiplying c_0 by c_5. To make these features transposition-invariant, the authors sum all cyclic shifts of the same interval in a given chroma vector $\mathbf{c}$. This calculation is simplified by applying a binary template $\mathbf{I}$ according to the desired type of interval or triad. A perfect fourth/fifth interval template is $\mathbf{I} = (1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)^\top$.

2.2 Tonal Complexity Features

Tonal Complexity features (denoted as $\mathbf{G}$) aim to capture musical attributes such as the amount of tonal variation in musical content. In [11], a total of seven complexity features are defined. For example, G_{CompEntr} represents the Shannon entropy of a chroma vector while $G_{\mathrm{CompFifth}}$ describes the spread of a chroma vector over the circle of fifths. Please refer to [11] for a thorough mathematical definition and musical interpretation of these features.

2.3 Tonal Interval Vectors

TIVs represent multi-level pitch in a geometrical space where vector distances relate to their perceived proximity [14]. The perceptual basis of the TIV space addresses three common limitations in preceding tonal pitch spaces [21–24]. First, it allows the representation and comparison of pitch at multiple time scales, namely individual pitches, chords, and keys. Second, prior knowledge of the key center is not required when measuring pitch distances. Third, it provides an indicator of consonance, lacking in related spaces. Moreover, distances between TIVs capture musical properties such as voice leading and shared interval content. Similar to the approach used in [25], the 6-dimensional complex TIV $\mathbf{T} \in \mathbb{C}^6$ is computed as the Discrete Fourier Transform (DFT) to a chroma vector $\mathbf{c} \in \mathbb{R}^N$

k	IC	Harmonic Quality	Intervals	w_k
1	IC1	Chromaticity	m2/M7	3
2	IC6	Dyadicity	Tritone	8
3	IC4	Triadicity	M3/m6	11.5
4	IC3	Diminished Quality	m3/M6	15
5	IC5	Diatonicity	P4/P5	14.5
6	IC2	Wholetoneness	M2/m7	7.5

Table 1. Harmonic quality and interval category associated with each TIV coefficient magnitude $\frac{\|T_k\|}{w_k}$.

as follows:

$$T_k = w_k \sum_{n=0}^{N-1} \bar{c}_n e^{-\frac{j2\pi kn}{N}}, \quad 1 \leq k \leq 6, \\ \text{with} \quad \bar{c}_n = \frac{c_n}{\sum_{n=0}^{N-1} c_n} \quad , \tag{1}$$

where $N = 12$ is the dimension of the chroma vector. We set $1 \leq k \leq 6$ due to the properties of the DFT by which the remaining coefficients are symmetric. $\mathbf{w} = (3, 8, 11.5, 15, 14.5, 7.5)$ are weights adjusting the contribution of each dimension T_k to improve the perceptual basis of the space in representing musical audio. They adjust the contribution of each coefficient T_k and promote the importance of the most relevant intervals within tonal music, such as fourths/fifths and major and minor thirds/sixths. These weights are derived from empirical dissonance ratings of complementary intervals and triads (major/minor, sus4, augmented, diminished). The adoption of the weights $\mathbf{w}$ have shown to capture perceptual distances between musical audio sonorities irrespective of timbral differences to a greater degree than chroma vectors or an unweighted DFT space [26].

2.4 TIV Basic Features

We now describe a group of features directly computed from TIVs, referred to as TIV Basic (denoted as $\mathbf{B}$). Musical interpretations are attributed to the magnitude $\|T_k\|/w_k$ of each of the six coefficients, evaluating the intervallic content of pitch configurations and its associated harmonic quality (see Table 1). For example, $k = 5$ corresponds to the $B_{\mathrm{Diatonicity}}$ feature. We establish a correspondence between each coefficient magnitude and the six complementary interval categories defined in Western music theory [20].

A TIV's indicator of consonance is computed as its magnitude (vector length) normalized to the norm of the weight vector $\mathbf{w}$, such that:

$$B_{\mathrm{Dissonance}} = 1 - \|\mathbf{T}\|/\|\mathbf{w}\| \quad . \tag{2}$$

3. FEATURE DESIGN IN THE TONAL INTERVAL SPACE

Aiming to take full advantage of the properties of TIVs, we now propose novel mid-level harmonic audio features that consider the harmonic structure and long-term hierarchical dependencies between audio segments.

3.1 TIV Complexity Features

Our first feature group, denoted as **Q**, captures aspects of tonal complexity and consonance, somehow mirroring the features presented in Section 2.2.

3.1.1 Distance Between Audio Segments

Distances between TIVs relate to their perceived similarity such that small distances equate to perceptually similar sonorities. To this end, Euclidean and cosine distance metrics between TIVs, which measure different perceptual characteristics of the signal, have been considered in the literature [14, 27].

The cosine distance θ measures the phase difference between two TIVs, capturing pitch class distances that roughly correspond to voice leading parsimony. Smaller values indicate a higher number of shared pitch classes.

The Euclidean distance metric d captures the phase φ_k and magnitude $\|T_k\|$ differences between TIVs. In addition to measuring the number of shared pitch classes between TIVs, it also accounts for shared interval content.

To exploit the musical properties of these two distance metrics, we propose new features that measure the perceptual distance between consecutive musical audio segments. Given the TIV $\mathbf{T}_n = (T_{0,n}, \ldots, T_{5,n})$ of the n-th segment, the Euclidean (d) and cosine (θ) distances between two consecutive segments is defined as:

$$\begin{aligned} Q_{\text{EucDist}} &= s_n^{\text{euc}} = d\{\mathbf{T}_n, \mathbf{T}_{n+1}\}, \\ Q_{\text{CosDist}} &= s_n^{\text{cos}} = \theta\{\mathbf{T}_n, \mathbf{T}_{n+1}\} \end{aligned} \tag{3}$$

The metrics can be applied at multiple hierarchies (or time scales) by adopting musical audio segments of different sizes. While short segments capture chord changes, larger segments can capture the overall tonal structure (e.g., key modulations).

3.1.2 Tonal Dispersion

The tonal dispersion of the n-th audio segment n is defined as the distance of its respective TIV to the tonal center $\overline{\mathbf{T}}$:

$$\begin{aligned} Q_{\text{EucTonalDisp}} &= u_n^{\text{euc}} = d\{\mathbf{T}_n, \overline{\mathbf{T}}\}, \\ Q_{\text{CosTonalDisp}} &= u_n^{\text{cos}} = \theta\{\mathbf{T}_n, \overline{\mathbf{T}}\} \end{aligned} \tag{4}$$

The tonal center of a piece corresponds to the TIV of the pitch-class distribution averaged across its entire duration. The tonal dispersion indicates how much the harmonic content of a given segment deviates from the tonal center $\overline{\mathbf{T}}$ and can indicate the degree of modulations across the piece or the segments with non-diatonic pitch classes (i.e., further away from the tonal center). This feature can help discriminate later classical musical periods that typically have larger tonal dispersion [28]. To compute the distance of a given segment n from the tonal center, we adopt both the Euclidean and cosine distance metrics to capture the different harmonic aspects mentioned in Section 3.1.1.

3.1.3 TIV Entropy

Related literature has been adopting Shannon entropy to capture the harmonic complexity within musical style [11, 29]. Amiot [30] recently proposed the computation of harmonic complexity from symbolic pitch-class distributions as the Shannon entropy of its Fourier coefficients. Following [30], we propose a TIV entropy feature to capture harmonic complexity from musical audio. The TIV entropy is high for random pitch-class distributions and low for pitch-class distributions that exhibit some degree of organization (i.e., periodicity or sparseness). We define the TIV entropy feature as the entropy of the normalized coefficient magnitudes of a TIV $\mathbf{T}$:

$$Q_{\text{TIVEntropy}} = \sum_{k=1}^{6} -p_k \log p_k, \quad p_k = \frac{\|T_k\|}{\sum_{j=1}^{6} \|T_j\|} \tag{5}$$

While entropy has been shown to capture different stylistic attributes depending on the time scale under analysis [29], we demonstrate in Table 2 the TIV entropy of various pitch-class sets. Lower TIV entropy (i.e., denoting greater organization) includes fairly common pitch-class distributions such as diatonic scales, triads, and tetrads. Higher TIV entropy denotes musical structures with less common adoption in the tonal music lexica, such as the chromatic set and its many subsets.

Pitch class set	TIV Entropy
Octatonic scale	0.3551
Diminished seventh chord	0.3552
Whole tone scale	0.5268
Diatonic scale	1.2444
Pentatonic scale	1.2972
Harmonic minor scale	1.4927
Minor seventh chord	1.5542
Single note	1.6767
Chromatic scale	1.6906

Table 2. TIV entropy of several pitch-class sets. For details on the elements of each set please refer to [30].

3.2 Harmonic Rhythm Features

Harmonic rhythm is the rate at which the chords change in a musical piece. It can provide a prominent indicator to distinguish style periods. For example, a fast harmonic rhythm is a characteristic of the Baroque period [31] in comparison with the large-scale structures of a Romantic symphony. To this end, we advance new features (denoted as **R**) that study the rate and magnitude of harmonic changes based on the Harmonic Change Detection Function (HCDF) proposed in [32]. The value ξ_n of the HCDF at frame n is defined as the rate of change between the TIVs $\mathbf{T}_{n-1}$ and $\mathbf{T}_{n+1}$ after Gaussian smoothing and is computed using the Euclidean distance.

Let us now consider Ξ_m as the set of frame indices of the peaks of the HCDF, considering all local maxima as peaks. We define the inter-peak interval feature as the difference between the frame numbers of consecutive peaks of the HCDF:

$$R_{\text{HCDFPeakInterval}} = \Delta_m = \Xi_{m+1} - \Xi_m \tag{6}$$

As an additional feature, we consider the magnitude of the peaks, $R_{\text{HCDFPeakMag}}$, given by ξ_Y, with $Y = \Xi_m$, which captures the degree of the harmonic changes.

3.3 Temporal Resolution of Tonal Interval Vectors

We detail two segmentation strategies for the feature extraction process: multiple fixed-size segment resolutions (Section 3.3.1) and variable-size segments resulting from the analysis of harmonic changes (Section 3.3.2).

3.3.1 Fixed-size Segmentation with Multiple Resolutions

Fixed-size segments with equal duration are adopted at multiple time scales or resolutions to capture the piece's different harmonic or tonal dimensions. Following [10–12], we consider four time resolutions in our work: 100ms, 500ms, 10s, and global (i.e., the entire piece or excerpt under analysis). Smaller time scales (e.g., 100ms and 500ms) capture finer musical elements such as individual notes, intervals, and chords. Larger time scales (e.g., 10s and global) capture higher harmonic structures at the level of structural parts or the overall piece, enhancing the harmonic changes in the large-scale tonal structure of the composition (such as key and modulations) and the harmonic trajectories of the horizontal structure (such as chord progressions).

3.3.2 Harmonic Structural Segmentation

We define a context-sensitive segmentation at the peaks of the HCDF (see Section 3.2). This strategy considers prominent harmonic changes as segment boundaries, therefore producing segments of varying duration according to the specific harmonic changes in the audio content. The use of context-sensitive segmentation in the feature extraction process of musical analysis systems has been shown to improve their accuracy compared to fixed segmentation approaches, which are less aware of musical structure [33, 34].

4. EXPERIMENTS AND RESULTS

To assess the degree to which the harmonic information conveyed by TIV harmonic audio features in Section 3 discriminate musical style, we evaluate the proposed features in two classical music style classification tasks from musical audio: historical periods (e.g., Baroque or Romantic) and composers. Furthermore, we compare the accuracy of the above features to the state-of-the-art template-based and complexity features described in Sections 2.1 and 2.2. To study the accuracy of different groups of features, we consider the two following sets: (template-based $\mathbf{F}$ and tonal complexity $\mathbf{G}$) and (TIV basic $\mathbf{B}$, TIV complexity $\mathbf{Q}$, harmonic rhythm $\mathbf{R}$). Finally, we inspect the impact of different context-sensitive vs. fixed-window segmentation strategies.

4.1 Experimental Procedure

For our experiments, we consider the *Cross-Era* [1] and *Cross-Composer* [2] datasets, which include 1600 and 1100 pieces, respectively. In detail, the *Cross-Era* dataset has 400 pieces per classical style period and features the following four periods: Baroque, Classical, Romantic, and Modern. The *Cross-Composer* dataset includes 100 pieces for each of the 11 featured classical music composers across all style periods. These datasets are further divided into the subsets presented in Table 3. The datasets provide pre-extracted NNLS chroma features at a resolution of 100ms (10Hz) for each piece.

Dataset	No. Classes (Z)	Items per Class
Cross-Era-Full	4	400
Cross-Era-Piano	4	200
Cross-Era-Orchestra	4	200
Cross-Comp-11	11	100
Cross-Comp-5	5	100

Table 3. Balanced subsets obtained from the *Cross-Era* and *Cross-Composer* datasets [12].

Based on [12], we employ the following classification procedure. Using the subsets listed in Table 3, we compute the features and calculate their piece-wise mean (μ) and standard deviation (σ). Then, we perform a stratified split into three cross-validation (CV) folds, one for testing and two others for training the model. Next, we use the two training folds to compute a Linear Discriminant Analysis (LDA) matrix to reduce the dimensionality of all three folds to $L = Z - 1$ with Z being the number of classes per task. We then perform a five-fold grid search on the two training folds to train an SVM classifier while optimizing its hyperparameters. We conduct this procedure three times, with each fold serving for testing once. To evaluate the robustness of the model concerning the random distribution into folds, we repeat the procedure ten times with re-initialized folds to calculate the accuracy deviation between the different classes (i.e., inter-class deviation).

To counteract problems stemming from adopting tracks from the same CD recording on the training and test folds (known as the album effect [35]), we take additional measures to prevent the model from adapting to the acoustic conditions of specific recordings. To this end, inspired by [11], we apply a composer filter (for period classification) or a performer filter (for composer classification) during the CV procedure that forces pieces from the same composer/performer to be placed within the same fold, thus avoiding overfitting while making the task more challenging and closer to real-world application scenarios.

4.2 Influence of Different Types of Segmentation

We analyze the impact of different audio segmentation strategies on the classification of classical music periods. Table 4 displays the classification accuracy for the TIV Basic ($\mathbf{B}$), TIV Complexity ($\mathbf{Q}$), and Harmonic Rhythm ($\mathbf{R}$) feature groups, as well as the combination of all the above features (through concatenation), on the *Cross-Era* dataset. As segmentation strategies, we consider fixed-size segmentation with a single resolution of 100ms (FS), fixed-size segmentation with multiple resolutions (MR)—100ms, 500ms, 10s, and global—and harmonic structural

[1] https://www.audiolabs-erlangen.de/resources/MIR/cross-era
[2] https://www.audiolabs-erlangen.de/resources/MIR/cross-comp

	FS	MR	HS	MR + HS
Cross-Era-Piano				
TIV Basic (B)	57.54%	**66.84%**	51.65%	65.64%
TIV Complexity (Q)	56.03%	**57.99%**	56.07%	57.67%
Harm. Rhythm (R)	-	-	21.26%	-
Combined	62.61%	**65.47%**	48.97%	63.80%
Cross-Era-Orchestra				
TIV Basic (B)	66.84%	74.80%	64.22%	**75.34%**
TIV Complexity (Q)	67.35%	69.87%	64.80%	**70.00%**
Harm. Rhythm (R)	-	-	28.31%	-
Combined	73.80%	77.19%	68.89%	**77.78%**
Cross-Era-Full				
TIV Basic (B)	60.41%	**70.13%**	57.24%	70.08%
TIV Complexity (Q)	55.97%	62.18%	57.35%	**62.86%**
Harm. Rhythm (R)	-	-	21.65%	-
Combined	64.94%	**71.63%**	62.30%	70.99%

Table 4. Classification accuracy for different types of segmentation: fixed-size (FS), fixed-window with multiple temporal resolutions (MR), harmonic structural segmentation (HS), and a combination of the last two approaches.

segmentation (HS). Additionally, we consider combining the two approaches presented in Section 3.3 (MR + HS).

For the *Cross-Era-Piano* and *Cross-Era-Full* datasets, the MR strategy shows slightly higher accuracy than other segmentation strategies. For other scenarios such as the *Cross-Era-Orchestra* dataset, combining the MR and HS approaches leads to similar or slightly better results. Overall, the MR strategy obtains the highest accuracy values in most cases and is less computationally expensive than the HS approach. Therefore, we opt for the MR strategy in all subsequent experiments, using the HS strategy only for the harmonic rhythm feature group $\mathbf{R}$.

4.3 Style Period and Composer Classification

Using the MR strategy, we perform style period and composer classification experiments on a larger set of feature combinations and show the results in Table 5. We first compare the template-based ($\mathbf{F}$) and TIV basic ($\mathbf{B}$) feature groups, which capture similar harmonic aspects. Indeed, we observe that these groups perform similarly on most datasets, with accuracy differences of 2.98% and 4.74% on the *Cross-Era-Piano* and *Cross-Comp-5* datasets, respectively. The tonal complexity ($\mathbf{G}$) and TIV complexity ($\mathbf{Q}$) groups show a greater performance accuracy difference in the *Cross-Era-Piano* dataset (7.85%) and a smaller difference in the *Cross-Era-Full* dataset (3.33%). Concerning the two largest datasets (*Cross-Era-Full* and *Cross-Comp-11*), the best feature groups result from combining TIV features with those proposed in [10, 11], achieving 74.04% and 38.25% accuracy, which correspond to an improvement of 2.88% and 4.74%, respectively, compared to using only the features proposed in previous work.

From these findings, we draw two main conclusions. First, conceptually similar feature groups lead to quite similar classification accuracies—an encouraging finding, which proves that our approach is valid (a kind of "sanity check" against [10, 11]) and indicates that the features exhibit the intended mid-level semantic properties (feature groups describing related harmonic properties behave similarly for classification). Second, the novel features at least partially capture complementary information such that their combination improves upon individual groups. Overall, we do not reach the state-of-the-art results on the *Cross-Era-Orchestra* and *Cross-Era-Full* subsets, which were obtained in [13] using chord transition features in combination with template-based and complexity features, surpassing our results by 6.01% and 4.16%. However, since TIV-based features improve upon the ones of [10,11], we hypothesize that, in combination with the chord transition features of [13], they might be able to reach even better classification accuracies.

4.4 Harmonic Features Correlation

We adopt hierarchical clustering to assess the degree of information (and redundancy) of harmonic audio features. Figure 1 shows the hierarchical clustering of all harmonic features detailed in Sections 2.2, 2.1, and 3, which we adopt in our experiments. Features are computed at a fixed-size 100ms resolution with average and standard deviation statistics.

Template-based ($\mathbf{F}$) and TIV basic ($\mathbf{B}$) features appear strongly correlated, often ending up in the same cluster. This suggests they may be capturing similar musical properties. For example, the two top-most features $F_{\mathrm{IC5},\sigma}$ and $B_{\mathrm{Diatonicity},\sigma}$ are direct neighbours and are both based on perfect-fifth relationships. Tonal Complexity ($\mathbf{G}$) and TIV Complexity ($\mathbf{Q}$) features also appear correlated, albeit to a slightly lower degree. For example, the green cluster is mainly composed of features from these groups. On the other hand, the red cluster contains several TIV Complexity features and none from the Tonal Complexity group, suggesting that those, in particular, describe different harmonic characteristics. The adopted global statistics, mean and standard deviation, tend to be grouped under the same cluster, suggesting to capture complementary information.

5. CONCLUSIONS AND FUTURE WORK

In this paper, we proposed a novel set of mid-level and perceptually-inspired harmonic audio features based on TIVs, which provide indicators for musical dissonance, chromaticity, dyadicity, triadicity, diminished quality, diatonicity, and whole-toneness, as well as quantify perceptual relations of short- and long-term harmonic structures, tonal dispersion, harmonic changes, and complexity. Features are computed using context-sensitive harmonic structure segmentation and a fixed-size segmentation with multiple temporal resolutions.

Proposed TIV harmonic features were assessed in two classical music style period and composer classification tasks using five different datasets. The novel TIV harmonic features have been compared to state-of-the-art harmonic features proposed in [10–12] and showed similar results, slightly outperforming in four out of the five datasets, with an improvement of up to 4.74%. A high degree of re-

		Cross-Era-Piano	Cross-Era-Orch	Cross-Era-Full	Cross-Comp-11	Cross-Comp-5
Tonal [10, 11]	**Template-based (F)**	**69.82 ±13.45**	75.25 ±12.16	70.51 ±11.57	37.41 ±19.96	50.01 ±19.77
	Tonal Complexity (G)	65.84 ±12.80	71.68 ±14.09	65.51 ±13.26	29.74 ±21.33	43.32 ±25.68
	Combined (F, G)	67.77 ±15.36	75.23 ±12.17	71.16 ±11.42	36.97 ±21.59	48.86 ±22.47
TIV	**TIV Basic (B)**	66.84 ±15.31	74.80 ±11.22	70.13 ±12.52	37.84 ±20.66	**54.75 ±17.96**
	TIV Complexity (Q)	57.99 ±17.67	69.87 ±12.87	62.18 ±13.71	29.61 ±19.77	43.16 ±19.06
	Harm. Rhythm (R)	21.26 ±28.06	28.31 ±21.51	21.65 ±26.44	7.77 ±17.64	16.47 ±19.35
	Basic + Comp. (B, Q)	65.59 ±16.77	76.68 ±10.35	71.50 ±12.03	37.82 ±20.30	53.40 ±16.85
	Combined (B, Q, R)	65.47 ±16.63	**77.19 ±9.89**	71.63 ±12.03	37.83 ±20.01	53.75 ±16.65
Combined	**F, G, B, Q, R**	64.39 ±16.27	76.70 ±11.41	73.78 ±10.80	**38.25 ±21.01**	49.72 ±19.25
Combined, no R	**F, G, B, Q**	64.78 ±15.80	76.56 ±11.29	**74.04 ±10.7**	37.89 ±21.57	50.44 ±18.72
Tonal+Transitions [13]		73.2	83.2	78.2	–	–

Table 5. Classification results for several feature groups across the *Cross-Era* and *Cross-Composer* datasets. The values on the table represent the mean classification accuracy and inter-class deviation, both expressed as percentages. For comparison, the state-of-the-art results for the *Cross-Era* dataset reported in [13] were also included.

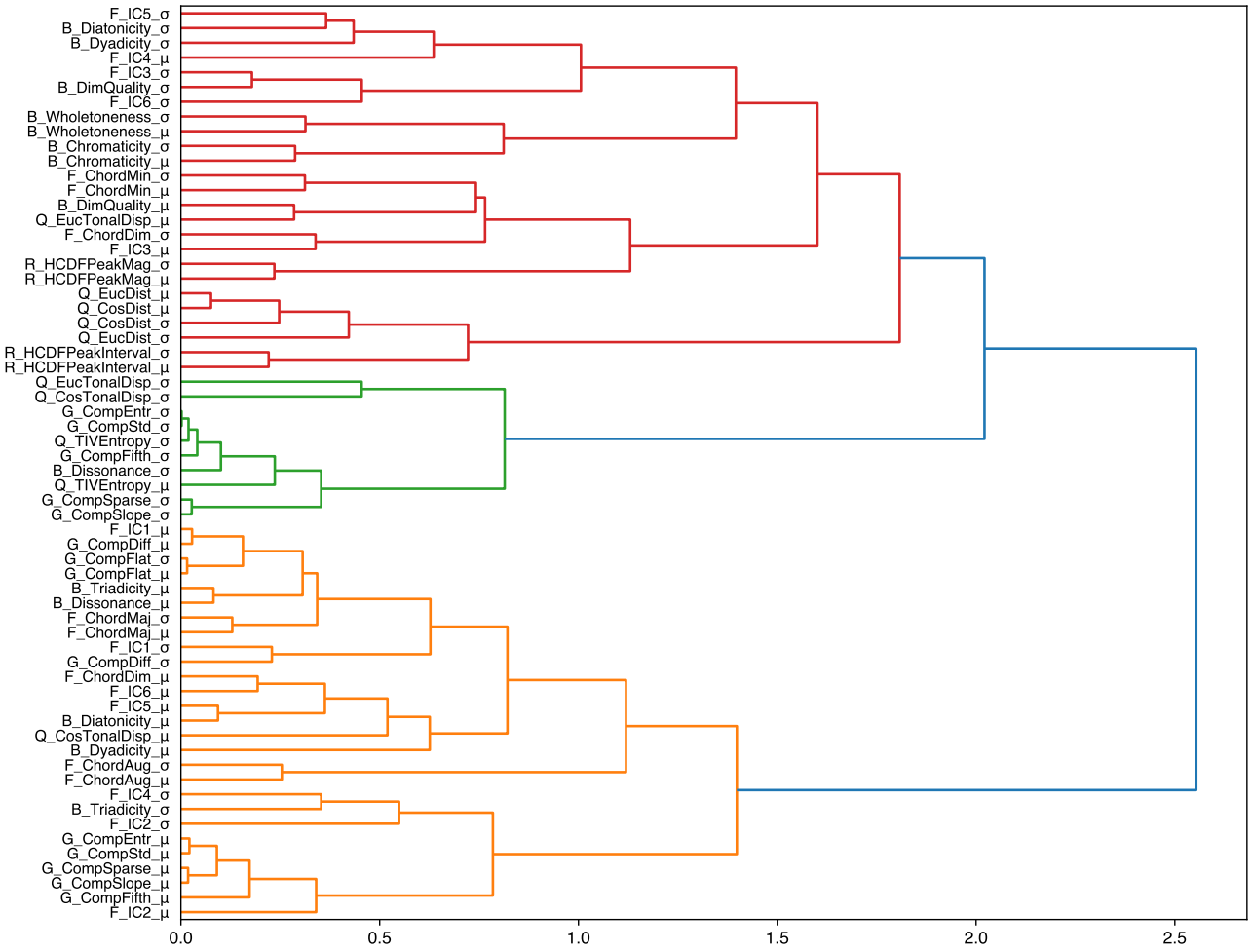


Figure 1. Hierarchical clustering of harmonic features computed at a 100ms resolution. The clustering is based on Spearman correlation distances using Ward's linkage method.

dundancy with existing state-of-the-art features has been found. However, the results suggest that new information is captured in the proposed TIV harmonic features, namely in what concerns the horizontal or long-term harmonic structure, e.g., tonal dispersion and distance between audio segments, and harmonic rhythm. While the context-sensitive segmentation strategy introduced slight improvements to classification accuracy, these do not seem to outweigh its higher computational cost.

Finally, for research reproducibility, we made the implementation of the proposed system publicly available online at `github.com/fcfalmeida/style-ident`. In future work, we may consider optimizing the segmentation strategy, expanding it to account for multiple time scales, conducting classification experiments on other musical genres, and experimenting with different machine learning approaches such as deep learning.

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